

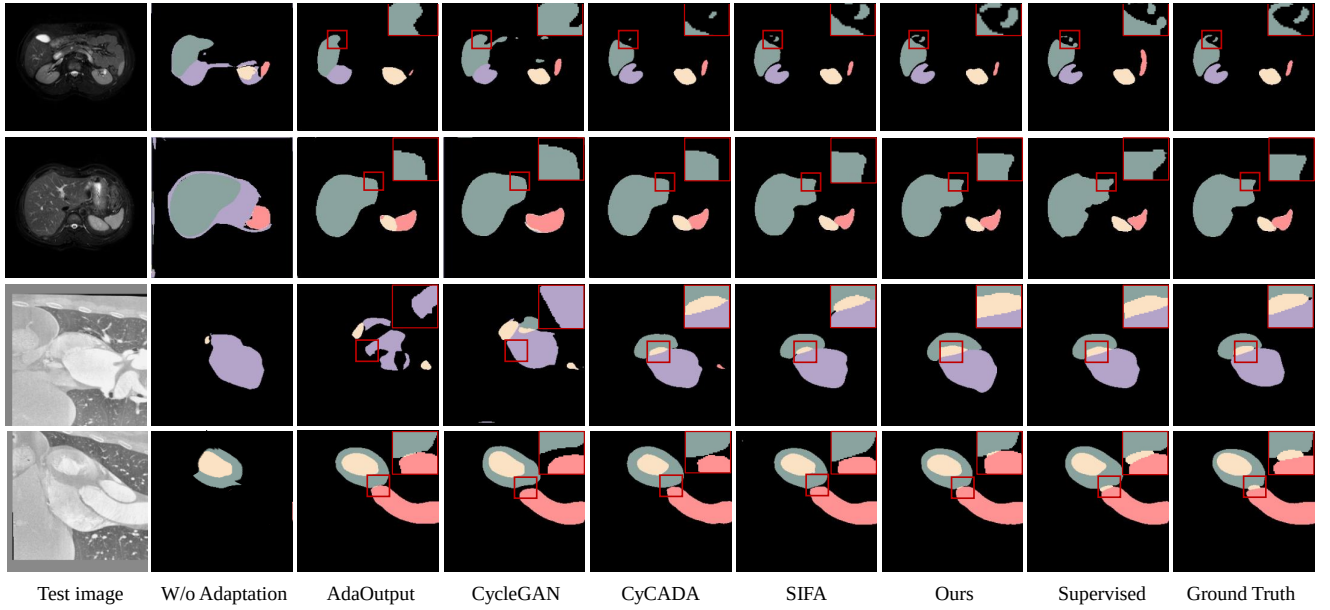


Figure 4: Visual comparison of segmentation results produced by different methods for abdominal multi-organ MRI images (the first and second rows) and cardiac CT images (the third and fourth rows). The liver, right kidney, left kidney and spleen are colored in blue, purple, yellow and red respectively. The cardiac structures of AA, LAC, LVC and MYO are colored in red, purple, yellow and blue respectively.

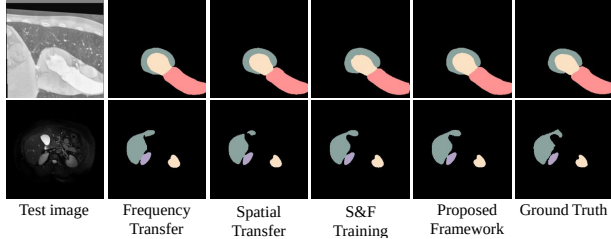


Figure 5: Visualization results of cardiac parts and abdominal organs using different settings in the ablation study.

5 Conclusion

In this paper, we propose a novel multi-teacher distillation framework for UDA based on frequency domain and spatial domain image transfer. In the frequency domain, we use NSCT to decompose images into 15 FCs and generate target-like images by combining selected FCs from the source and the target domain images for model training. In the spatial domain, we utilize histogram matching based on batch momentum update strategy to generate another set of target-like images for model training. Two teacher models are obtained with the frequency domain transferred images and the spatial domain transferred images, and a student model is then trained for inference under a multi-teacher distillation framework. Comparisons to existing methods and ablation studies validate the effectiveness of our proposed method. The idea of combining spatial and frequency domain transfer strategies in a multi-teacher distillation framework have the potential to be used in other UDA tasks.

Table 4: Experimental results of different histogram matching strategies on cardiac datasets (source domain: MRI, target domain: CT)

Method	Dice	ASD
Entire target domain images	81.5	3.4
Single target image	82.8	3.1
Batch average-based	83.4	3.3
Batch momentum update-based	83.7	3.2

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